Binary Data and Spread-Spectrum Communications

42057 Space Communications and Sensing — Assignment 2, Topic 2

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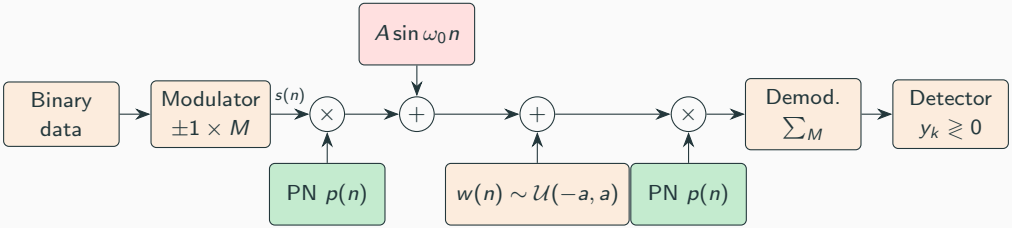
Why spread spectrum in space links? - Ben

- Space links are power limited and exposed to interference and jamming.
- GPS spreads a 50 bit/s message with a 1.023 Mchirps/s code: 43 dB processing gain.
- TDRSS uses direct-sequence spreading for multiple access and interference rejection.

Our task

1. Simulate a binary system in uniform noise and measure BER vs SNR.
2. Add a PN sequence and a sinusoidal jammer $A \sin \omega_0 n$, $A \geq M$, and compare with and without PN for $M = 50, 100, 500, 1000$.

System models - Ben



Project 12.7: green and red blocks removed.

$y_k = \pm M + \sum w(n), \quad \mathrm{SNR} = M/P_n,$
$P_n = a^2/3.$

Project 12.8: full diagram.

Same PN at both ends, $p^2(n) = 1$, so the signal is restored exactly.

MATLAB implementation - Ben

One function per block, reused across both projects:

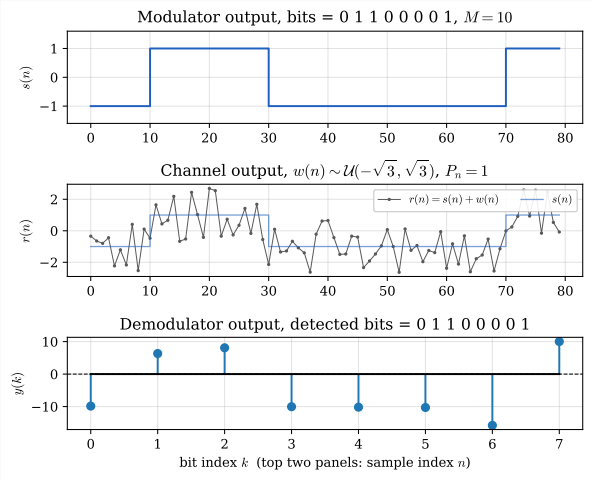
- `bit_generator`, `modulator`
- `noise_generator`, `demodulator`
- `detector` (and error counter)
- `pn_generator`, `sine_generator`
- `binary_comm_sim`: Monte-Carlo loop

No toolboxes; block-wise processing for $M = 1000$;
run until 200 errors or 2×10^7 bits.

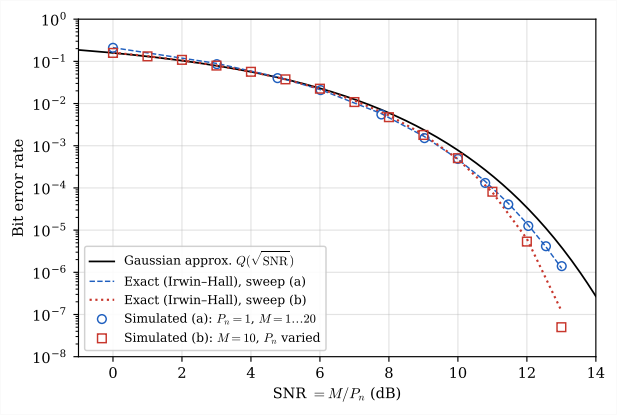
Core loop

```
b = bit_generator(Nb);
s = modulator(b, M);
if use_pn, p = pn_generator(L); s = s.*p; end
r = s + noise_generator(L, a) ...
      + sine_generator(L, A, w0, n0);
if use_pn, r = r.*p; end
y = demodulator(r, M);
[~, e] = detector(y, b);
```

Project 12.7: signals through the system - Elina



Project 12.7: BER vs SNR - Elina



- Simulation matches the exact closed-form (Irwin–Hall) result down to 10^{-7}.
- $\mathcal{Q}(\sqrt{\mathrm{SNR}})$ is pessimistic at high SNR: uniform noise is bounded.
- Sweeps diverge at high SNR: with $M = 10$ fixed, $a \to 1$ and errors become impossible ($Ma \leq M$).

Theory: what the PN sequence does to the jammer - Elina

Without PN

$$J_k = A\,\frac{\sin(M\omega_0/2)}{\sin(\omega_0/2)}\sin(\cdot)$$

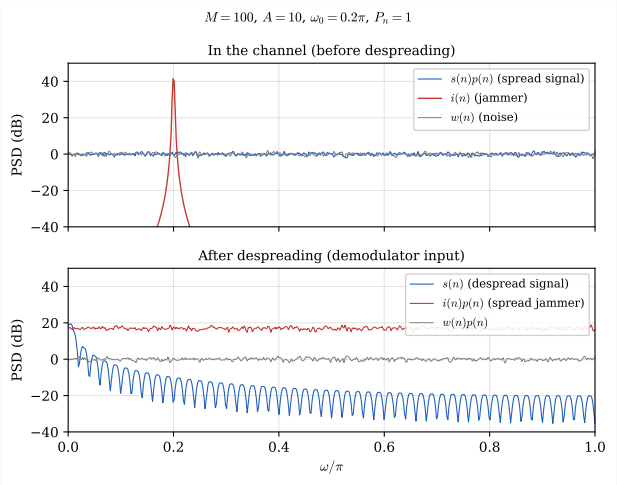
- In-band ($\omega_0 \ll 2\pi/M$): $|J_k| \to AM$, integrated like the signal.
- Out-of-band: $|J_k| \le A/\sin(\omega_0/2)$, already filtered.

With PN

$$\mathrm{Var}[J'_k] \approx \frac{MA^2}{2}, \quad P_e \approx \mathcal{Q}\left(\sqrt{\frac{M}{P_n + A^2/2}}\right)$$

- Jammer becomes white noise of power $A^2/2$.
- Processing gain $G_p = M$ (17 dB to 30 dB).
- But $A = M$ means $\mathrm{JSR} = M^2/2$: SINR $\approx 2/M$.

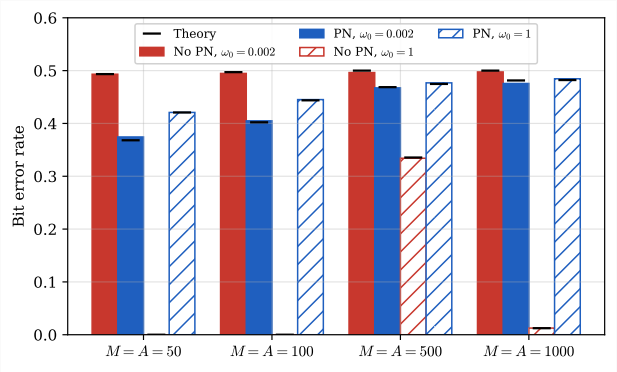
Project 12.8: spectra before and after despreading - Rhyse



$M = 100,\ A = 10,\ \omega_0 = 0.2\pi$

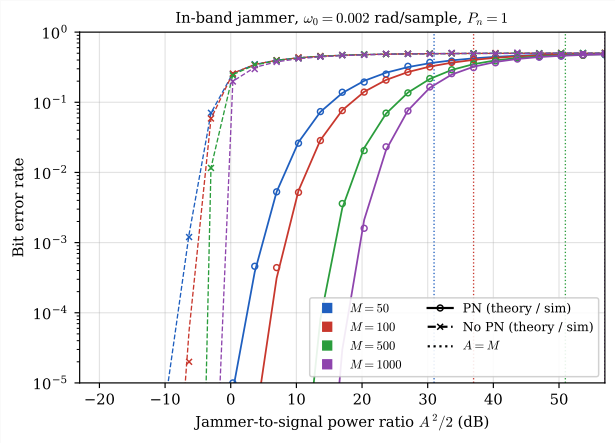
- Channel: signal is white at 0 dB, jammer is a 40 dB line.
- After despreading: signal returns to a sinc^2 with 20 dB at DC; jammer is a white floor at 17 dB.
- The demodulator passes only $|\omega| \lesssim 2\pi/M$, about $1/M$ of the jammer power.

Project 12.8: the brief's experiment, $A = M$ - Rhyse



- **In-band**: no PN ≈ 0.5; PN 0.37 at $M = 50$, rising to 0.48.
- **Out-of-band**: no PN is often *better*; the integrator rejects the tone.
- Theory (black ticks) matches within 0.007.
- JSR of 31 dB to 57 dB exceeds the processing gain.

Processing gain: BER vs jammer power - Rhyse

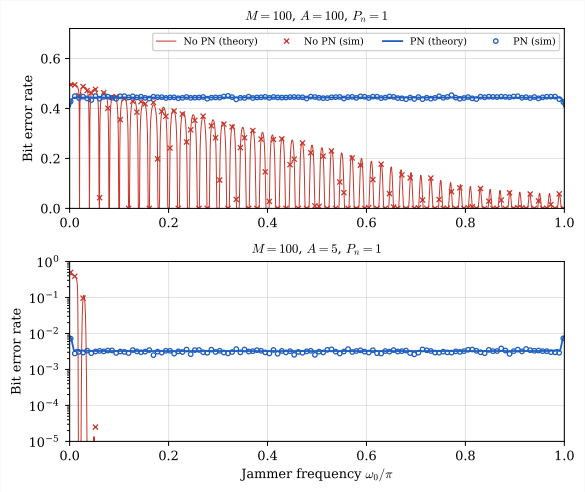


JSR tolerated at BER $= 10^{-3}$:

M	No PN	PN	Gain
50	−6.5	4.9	11.4 dB
100	−5.0	8.3	13.3 dB
500	−3.3	16.0	19.2 dB
1000	−0.7	19.7	20.5 dB

Without PN, more samples per bit do not help: the jammer is integrated too.

BER vs jammer frequency ($M = 100$) - Jaylen



- PN: BER is nearly flat in frequency; it changes only within $\sim 2\pi/M$ of 0 and π.
- No PN: BER follows the Dirichlet kernel, zero at the nulls $2\pi\ell/M$, worst near DC.
- A jammer can always pick the conventional receiver's weak spot: the signal band.

Why spread spectrum wins (and when it doesn't) - Jaylen

1. **Coherent vs incoherent addition**: the signal sums to M (power M^2), the despread jammer to power $MA^2/2$, a gain of M.
2. **Weak frequency dependence**: away from the band edges there is no weak frequency to exploit.
3. **Low probability of intercept and CDMA**: the signal sits at the noise floor.

Caveat from the $A \geq M$ condition

The jammer power grows as M^2 but the processing gain only as M, so both systems sit near BER $= 0.5$. The benefit of spreading is a fixed number of decibels.

Conclusions - Jaylen

- Modular MATLAB simulator, one function per block, validated against exact theory.
- 12.7: BER matches the exact uniform-noise result; the Gaussian approximation is pessimistic at high SNR.
- 12.8: PN despreading turns a tone into white noise, giving 11 dB to 20 dB more jamming tolerance for $M = 50$ to 1000 and nearly frequency-independent performance.
- With $A = M$ the jammer exceeds the processing gain by 14 dB to 27 dB.
- Limitations: perfect synchronisation; random PN instead of m-sequences or Gold codes.

Questions?